

Climate 5000 Series - Gen 4

9K BTU Single Zone Max Performance System

4-Way Cassette IDU

Engineering
Submittal
Sheet



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Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

9K Btu Single Zone Max Performance System - 4-Way Cassette IDU

Indoor part number 8-733-962-716

Indoor model number BMS500-AAU009-1AHCXD

Outdoor part number 8-733-962-688

Outdoor model number BMS500-AAS009-1CSXHD

Efficiency Data AHRI 210/240 2023

SEER2 22.6

EER2 14.0

HSPF2 Region IV 12.1

HSPF2 Region V 9.3

COP(47F) 4.00

COP (5F) 1.84

Performance Data AHRI 210/240 2023

Cooling	Capacity Range	3700~13600
	Rated Capacity	9000
	Current	4.8
	Input Power	643

Heating	Capacity Range	5100~15700
	Rated Capacity (47F)	10000
	Rated Capacity (5F)	11000
	Current	3.3
	Input Power	733

Electrical Data

Power supply (V / Ø / Hz) 208-230 / 1 / 60

Min. circuit ampacity indoor/outdoor (A) 3.0 / 15.0

Max. fuse indoor/outdoor (A) 15.0 / 15.0

Compressor Data

Type: Rotary

Refrigerant oil /oil charge (ml) VG74 / 350

Input (W) 885

Rated current (RLA/A) 6.0

Piping

Refrigerant precharge (oz/kg) 35.27 / 1.00

Length for precharge (ft/m) 24.6 / 7.5

Liquid side (in/mm) 1/4" / Ø6.35

Gas side (in/mm) 3/8" / Ø9.52

Additional Charge of Refrigerant (oz/ft)/(g/m) 0.16 / 15

Max. refrigerant pipe length (ft/m) 82.00 / 25

Max. difference in height (ft/m) 49.20 / 15

Operation Range

ODU	Cooling (°F / °C)	-22~-122 / -30~50
	Heating (°F / °C)	-22~-75 / -30~-24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		1.86
IDU	Fan motor model	ZKFN-45-8-2
	Fan motor qty (ea)	1
	Fan motor input (W)	50.0
	Fan speed (RPM)	600 / 530 / 380
	Air flow, Turbo / H / M / L / Si (CFM)	353.2 / 312.0 / 264.9 / 182.5 / 147.2
	Approx. Air Throw Range ft (m)	9.8 / 3
	Noise level, Turbo / H / M / L / Si (dB(A))	42.5 / 40.0 / 37.5 / 26.0 / 24.0
	Net weight (lbs / kg)	35.27 / 16.0
	Gross weight (lbs / kg)	41.89 / 19.0
	Panel net weight (lbs / kg)	5.95 / 2.7
ODU	Gross panel weight (lbs / kg)	9.48 / 4.3
	Drain pipe diameter (mm)	ODΦ25mm
	Fan motor model	ZKFN-34-10-1-3
	Fan motor qty (ea)	1
	Fan motor input (W)	50
	Fan motor speed (RPM)	850 / 680 / 450
	Air flow, Max (CFM)	1471.5
	Noise level, Max (dB(A))	55.0
	Net weight (lbs / kg)	72.75 / 33.0
	Gross weight (lbs / kg)	78.7 / 35.7

Accessories

BMS-CCG-XXD (8733962708)	Decorative Grille - 9K ~ 18K
BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK1-7X1 (8733951010)	25ft Lineset

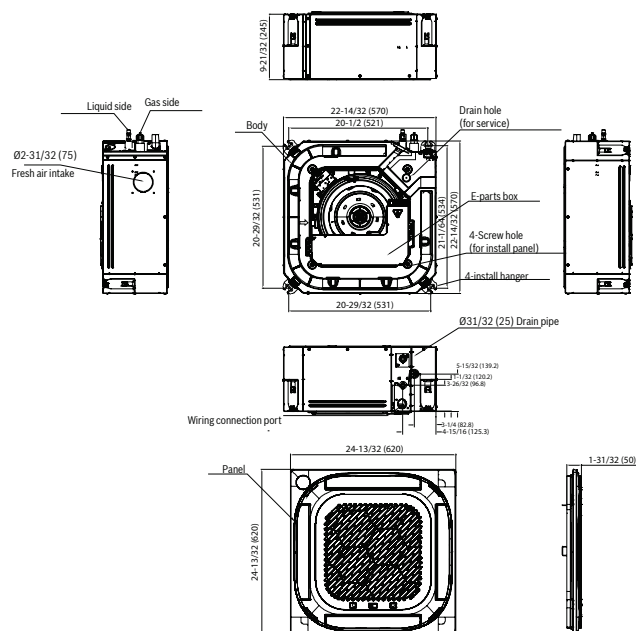
Standard Parts

8-733-969-065	Remote Control
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9K Btu Single Zone Max Performance System - 4-Way Cassette IDU

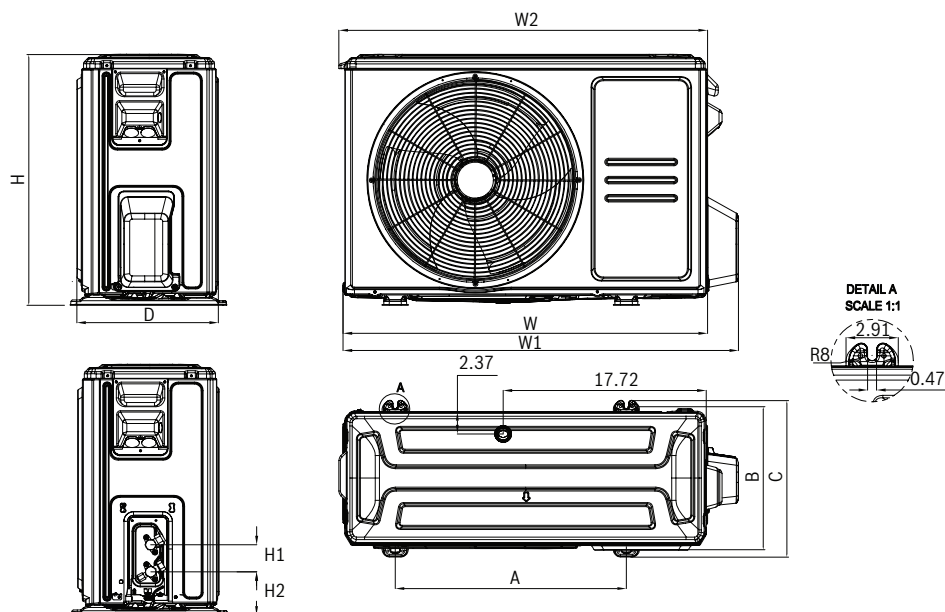


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU009-1AHCXD	22.44	22.44	9.65

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)	H1 (in)	H2 (in)
BMS500-AAS009-1CSXHD	31.69	12.99	21.81	34.38	32.13	20.13	12.5	13.63	2.38	3.75

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9K Btu Single Zone Max Performance System - 4-Way Cassette IDU


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Cooling Performance - Indoor Dry Bulb Temperature at 353 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	8523	8535	8544	8544	8556	9037	9048	9057	9058	9069	9577	9589	9600	9602	9613	10154	10165	10176	10178	10190
	S/T	0.60	0.71	0.80	0.80	0.91	0.57	0.67	0.76	0.76	0.86	0.53	0.63	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.77
	PI	0.33	0.33	0.33	0.33	0.33	0.35	0.35	0.35	0.35	0.35	0.37	0.37	0.37	0.37	0.37	0.39	0.39	0.39	0.39	0.39
-13°F (-25.0°C)	TC	9234	9246	9256	9256	9269	9790	9802	9811	9813	9825	10375	10388	10400	10402	10414	11000	11012	11024	11026	11039
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.68	0.68	0.78
	PI	0.34	0.34	0.34	0.34	0.34	0.36	0.36	0.36	0.36	0.36	0.38	0.38	0.38	0.38	0.38	0.40	0.40	0.40	0.40	0.40
-4°F (-20.0°C)	TC	9944	9957	9968	9968	9982	10543	10556	10566	10568	10581	11173	11188	11200	11202	11215	11846	11859	11872	11874	11889
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.79
	PI	0.35	0.35	0.35	0.35	0.35	0.37	0.37	0.37	0.37	0.37	0.39	0.39	0.39	0.39	0.39	0.42	0.42	0.42	0.42	0.42
0°F (-17.8°C)	TC	10260	10273	10284	10285	10299	10878	10891	10901	10904	10917	11528	11543	11556	11558	11571	12222	12236	12249	12251	12266
	S/T	0.63	0.73	0.84	0.84	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	0.36	0.36	0.36	0.36	0.36	0.38	0.38	0.38	0.38	0.38	0.40	0.40	0.40	0.40	0.40	0.42	0.42	0.42	0.42	0.43
5°F (-15.0°C)	TC	10654	10668	10680	10680	10695	11296	11310	11321	11323	11336	11971	11987	12000	12002	12016	12692	12706	12720	12722	12738
	S/T	0.63	0.74	0.84	0.84	0.96	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81
	PI	0.36	0.36	0.36	0.36	0.36	0.38	0.38	0.38	0.38	0.39	0.41	0.41	0.41	0.41	0.41	0.43	0.43	0.43	0.43	0.43
14°F (-10.0°C)	TC	11574	11590	11602	11603	11619	12272	12286	12298	12301	12316	13005	13022	13036	13039	13053	13789	13804	13819	13821	13838
	S/T	0.62	0.73	0.83	0.83	0.94	0.59	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79
	PI	0.39	0.39	0.39	0.39	0.40	0.42	0.42	0.42	0.42	0.42	0.44	0.44	0.44	0.44	0.44	0.47	0.47	0.47	0.47	0.47
17°F (-8.3°C)	TC	11881	11897	11910	11910	11926	12597	12612	12624	12627	12642	13350	13367	13382	13384	13399	14154	14169	14185	14187	14205
	S/T	0.62	0.72	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.88	0.55	0.64	0.73	0.73	0.83	0.52	0.61	0.69	0.69	0.79
	PI	0.40	0.40	0.40	0.40	0.41	0.43	0.43	0.43	0.43	0.43	0.45	0.45	0.45	0.45	0.46	0.48	0.48	0.48	0.48	0.48
23°F (-5.0°C)	TC	11743	11759	11772	11772	11788	12451	12466	12478	12480	12495	13195	13212	13227	13229	13244	13990	14005	14020	14022	14040
	S/T	0.61	0.72	0.81	0.81	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78
	PI	0.43	0.43	0.43	0.43	0.43	0.45	0.45	0.46	0.46	0.46	0.48	0.48	0.48	0.48	0.48	0.51	0.51	0.51	0.51	0.51
35°F (1.7°C)	TC	11467	11483	11495	11496	11511	12159	12173	12185	12187	12202	12885	12902	12916	12918	12933	13661	13676	13691	13693	13710
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	0.48	0.48	0.48	0.48	0.48	0.51	0.51	0.51	0.51	0.51	0.54	0.54	0.54	0.54	0.54	0.57	0.57	0.57	0.57	0.57
41°F (5.0°C)	TC	11691	11707	11719	11720	11736	12395	12410	12422	12425	12440	13136	13153	13168	13170	13185	13927	13943	13958	13960	13977
	S/T	0.61	0.72	0.82	0.82	0.93	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.73	0.73	0.83	0.51	0.60	0.69	0.69	0.78
	PI	0.48	0.48	0.48	0.48	0.49	0.51	0.51	0.51	0.51	0.51	0.54	0.54	0.54	0.54	0.54	0.58	0.58	0.58	0.58	0.58
47°F (8.3°C)	TC	11914	11930	11943	11944	11960	12632	12647	12660	12662	12677	13387	13404	13419	13422	13437	14194	14209	14225	14227	14244
	S/T	0.63	0.73	0.83	0.83	0.95	0.59	0.69	0.79	0.79	0.90	0.56	0.65	0.74	0.74	0.85	0.53	0.62	0.70	0.70	0.80
	PI	0.49	0.49	0.49	0.49	0.49	0.52	0.52	0.52	0.52	0.52	0.55	0.55	0.55	0.55	0.55	0.58	0.58	0.58	0.58	0.59
50°F (10.0°C)	TC	11870	11886	11899	11899	11915	12585	12600	12613	12615	12630	13337	13355	13370	13372	13387	14141	14156	14172	14174	14192
	S/T	0.62	0.73	0.82	0.82	0.94	0.58	0.68	0.78	0.78	0.89	0.55	0.65	0.73	0.73	0.84	0.52	0.61	0.69	0.69	0.79
	PI	0.50	0.50	0.50	0.50	0.50	0.53	0.53	0.53	0.53	0.53	0.56	0.56	0.56	0.56	0.56	0.59	0.59	0.59	0.59	0.59
59°F (15.0°C)	TC	11737	11753	11766	11766	11782	12445	12459	12472	12474	12489	13188	13205	13220	13222	13237	13983	13998	14013	14015	14033
	S/T	0.60	0.70	0.80	0.80	0.91	0.56	0.66	0.75	0.75	0.86	0.53	0.62	0.71	0.71	0.81	0.50	0.59	0.67	0.67	0.76
	PI	0.51	0.51	0.51	0.51	0.51	0.54	0.54	0.54	0.54	0.54	0.58	0.58	0.58	0.58	0.58	0.61	0.61	0.61	0.61	0.61
68°F (20.0°C)	TC	11604	11620	11633	11633	11649	12304	12318	12331	12333	12348	13039	13056	13070	13073	13087	13825	13840	13855	13857	13874
	S/T	0.58	0.68	0.77	0.77	0.88	0.54	0.64	0.72	0.72	0.83	0.51	0.60	0.68	0.68	0.78	0.48	0.57	0.65	0.65	0.74
	PI	0.53	0.53	0.53	0.53	0.53	0.56	0.56	0.56	0.56	0.56	0.59	0.59	0.60	0.60	0.60	0.63	0.63	0.63	0.63	0.63
77°F (25.0°C)	TC	11472	11487	11500	11500	11516	12163	12177	12189	12192	12206	12890	12906	12921	12923	12938	13666	13681	13696	13698	13715
	S/T	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.70	0.70	0.80	0.49	0.58	0.66	0.66	0.75	0.47	0.55	0.62	0.62	0.71
	PI	0.54	0.55	0.55	0.55	0.55	0.58	0.58	0.58	0.58	0.58	0.61	0.61	0.61	0.61	0.61	0.65	0.65	0.65	0.65	0.65
82°F (27.8°C)	TC	11511	11526	11539	11539	11555	12204	12219	12231	12233	12248	12934	12950	12965	12967	12982	13713	13728	13743	13745	13762
	S/T	0.56	0.66	0.75	0.75	0.85	0.53	0.62	0.70	0.70	0.80	0.50	0.58	0.66	0.66	0.76	0.47	0.55	0.63	0.63	0.71
	PI	0.61	0.61	0.61	0.61	0.61	0.65	0.65	0.65	0.65	0.65	0.69	0.69	0.69	0.69	0.69	0.73	0.73	0.73	0.73	0.73
86°F (30.0°C)	TC	11684	11700	11713	11713	11729	12388	12403	12415	12418	12433	13129	13146	13160	13163	13177	13920	13935	13950	13952	13969
	S/T	0.58	0.69	0.78	0.78	0.89	0.55	0.65	0.74	0.74	0.84	0.52	0.61	0.69	0.69	0.79	0.49	0.58	0.65	0.65	0.75
	PI	0.74	0.74	0.74	0.74	0.74	0.78	0.78	0.78	0.78	0.78	0.83	0.83	0.83	0.83	0.83	0.88	0.88	0.88	0.88	0.88
95°F (35.0°C)	TC	12075	12091	12104	12104	12121	12802	12818	12830	12833	1284										

Climate 5000 Series - Gen 4

9K Btu Single Zone Max Performance System - 4-Way Cassette IDU



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Heating Performance - Indoor Dry Bulb Temperature at 353 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	4307	4221	4200	4183	4146	1.10	1.11	1.11	1.11	1.12
-13°F(-25.0°C)	5640	5528	5500	5478	5429	1.21	1.22	1.22	1.22	1.23
-4°F(-20.0°C)	7589	7437	7400	7370	7304	1.42	1.43	1.44	1.44	1.44
0°F(-17.8°C)	9230	9045	9000	8964	8883	1.56	1.57	1.58	1.58	1.59
5°F(-15.0°C)	11281	11055	11000	10956	10857	1.73	1.75	1.75	1.76	1.76
14°F(-10.0°C)	12512	12261	12200	12151	12042	1.74	1.75	1.76	1.76	1.77
17°F(-8.3°C)	12922	12663	12600	12550	12437	1.74	1.75	1.76	1.76	1.77
22°F(-5.6°C)	13504	13234	13168	13115	12997	1.72	1.73	1.74	1.74	1.75
27°F(-2.8°C)	14086	13805	13736	13681	13558	1.70	1.71	1.71	1.72	1.72
32°F(0.0°C)	14669	14375	14304	14246	14118	1.67	1.69	1.69	1.69	1.70
35°F(1.7°C)	15018	14718	14644	14586	14454	1.66	1.67	1.68	1.68	1.69
37°F(2.8°C)	15250	14945	14870	14811	14677	1.60	1.62	1.62	1.62	1.63
42°F(5.6°C)	15829	15513	15435	15373	15235	1.46	1.47	1.47	1.48	1.48
44.6°F(7.0°C)	16061	15740	15661	15598	15458	1.40	1.41	1.42	1.42	1.43
47°F(8.3°C)	16409	16080	16000	15936	15793	1.32	1.33	1.33	1.33	1.34
52°F(11.1°C)	15970	15650	15572	15510	15370	1.17	1.18	1.18	1.18	1.19
57°F(13.9°C)	15531	15220	15144	15084	14948	1.02	1.03	1.04	1.04	1.04
59°F(15.0°C)	15356	15049	14973	14913	14779	0.97	0.97	0.98	0.98	0.98
62°F(16.7°C)	15092	14791	14717	14658	14526	0.88	0.89	0.89	0.89	0.89
64.4°F(18.0°C)	14882	14584	14511	14453	14323	0.81	0.82	0.82	0.82	0.82
69.8°F(21.0°C)	14408	14120	14049	13993	13867	0.65	0.66	0.66	0.66	0.66
75.2°F(24.0°C)	13934	13655	13587	13533	13411	0.49	0.50	0.50	0.50	0.50

Bolded values were tested. All other values are simulated and for reference only.